Azure Redis Cache in ASP.NET Core

Azure Redis Cache Pricing Analysis

Implementing Redis Cache in ASP.NET core

Steps to integrate Redis cache into an ASP.NET Core application

- **Install Required NuGet Packages**
  Install the 'stackexchange.Redis' package for redis connectivity
- **Configure Redis**
  Add your Redis configuration details to appsettings.json.
- **Set up Redis in program.cs**
  In the Program.cs class, the Redis cache service is configured using the ConnectionMultiplexer extension method within the ConfigureServices section.

Configure Redis settings in the Azure portal

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➔ Create Redis Cache in Azure

1) Go to the Azure portal.
2) Search for "Azure Cache for Redis" and click "Create."
3) Configure basic settings, choose a pricing tier, and create the instance.

➔ Configure Redis cache

1) Access the Redis instance from the portal.
2) Copy the connection string from the "Access keys" tab.
3) Optionally configure networking, scaling, and diagnostics.

➔ Connect in ASP.NET Core

1) Go to the "Access keys" tab in your Redis instance, and copy the primary connection string.
2) Add the connection string in “Appsettings.json”.

➔ Run and Test Your Application

1) Test your application to ensure that it interacts with Redis as expected.
2) Use tools like redis-cli or a Redis GUI client to monitor and verify the data stored in Redis.

Azure Cache for Redis pricing- Basic

Cache Name	Cache Size	Network performance	Number of Client Connections	Price
C0	250 MB	Low	256	₹1.843/hour
C1	1 GB	Low	1,000	₹4.606/hour
C2	2.5 GB	Low	2,000	₹7.537/hour
C3	6 GB	Moderate	5,000	₹15.073/hour
C4	13 GB	Low	10,000	N/A
C5	26 GB	Moderate	15,000	₹35.171/hour
C6	53 GB	High	20,000	₹70.341/hour

Standard

Cache Name	Cache Size	Network performance	Number of Client Connections	Price (two-node)
C0	250 MB	Low	256	₹4.606/hour
C1	1 GB	Low	1,000	₹11.556/hour
C2	2.5 GB	Low	2,000	₹18.758/hour
C3	6 GB	Moderate	5,000	₹37.683/hour
C4	13 GB	Low	10,000	₹43.880/hour
C5	26 GB	Moderate	15,000	₹87.926/hour
C6	53 GB	High	20,000	₹175.852/hour

Premium

Cache Name	Cache Size	Network performance	Number of Client Connections	Pay As You Go (Per Shard, Two Node)	1 Year Reserved (% Savings)	3 Year Reserved (% Savings)	Pay as You Go Price per additional node
P1	6 GB	Moderate	7,500	₹60.460/hour	₹38.601/hour ~36% savings	₹27.142/hour ~55% savings	₹30.230/hour
P2	13 GB	High	15,000	₹120.752/hour	₹77.335/hour ~36% savings	₹54.374/hour ~55% savings	₹60.376/hour
P3	26 GB	High	30,000	₹241.503/hour	₹154.535/hour ~36% savings	₹108.657/hour ~55% savings	₹120.752/hour
P4	53 GB	Highest	40,000	₹483.341/hour	₹309.336/hour ~36% savings	₹217.505/hour ~55% savings	₹241.671/hour
P5	120 GB	Highest	40,000	₹1,093.629/hour	₹700.194/hour ~36% savings	₹492.319/hour ~55% savings	₹546.815/hour

Enterprise

Cache Name	Cache Size	Network performance	Number of Client Connections	Infrastructure Cost (pay as you go, two data node)	Software IP Cost (pay as you go, two data node)	Total Cost (pay as y go, two data node)
E1*	1 GB	Moderate	16,667	₹8.709/hour	₹3.400/hour	₹12.109/hour
E5	4 GB	Moderate	16,667	₹55.603/hour	₹15.743/hour	₹71.346/hour
E10	12 GB	High	50,000	₹104.724/hour	₹31.319/hour	₹136.042/hour
E20	25 GB	High	50,000	₹208.929/hour	₹62.470/hour	₹271.398/hour
E50	50 GB	Highest	1,00,000	₹410.404/hour	₹122.762/hour	₹533.165/hour
E100	100 GB	Highest	2,00,000	₹871.051/hour	₹245.523/hour	₹1,116.573/hour
E200	200 GB	Highest	2,00,000	₹1,742.101/hour	₹490.877/hour	₹2,232.978/hour
E400	400 GB	Highest	4,00,000	₹3,484.370/hour	₹981.754/hour	₹4,466.123/hour

Thank You

For the most accurate and up-to-date pricing, visit the Azure Redis Cache Pricing page